

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I Jabaraj T (192511187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

(a) Propositional Logic Representation

Let:

- **F** = Patient has Fever
- **C** = Patient has Cough
- **R** = Patient has Rash
- **B** = Patient has Breathlessness
- **Flu** = Patient has Flu
- **M** = Patient has Measles
- **P** = Patient has Pneumonia

The rules can be represented as:

1. **Fever AND Cough $\rightarrow$ Flu**

$F \wedge C \rightarrow \text{Flu}$

2. **Fever AND Rash $\rightarrow$ Measles**

$F \wedge R \rightarrow M$

3. **Cough AND Breathlessness $\rightarrow$ Pneumonia**

$C \wedge B \rightarrow P$

Facts about Patient A:

- **F = True**
- **C = True**
- **B = True**

Therefore, the initial facts are:

$F \wedge C \wedge B$

(b) Forward Chaining

Forward chaining starts with the known facts and applies rules whose conditions are satisfied.

Step 1

Known facts:

Fever, Cough, Breathlessness

Rule 1:

Fever $\wedge$ Cough $\rightarrow$ Flu

Since Fever and Cough are true:

Flu is derived.

Step 2

Rule 2:

Fever $\wedge$ Rash $\rightarrow$ Measles

Rash is not given as a fact.

Therefore:

Measles cannot be derived.

Step 3

Rule 3:

Cough $\wedge$ Breathlessness $\rightarrow$ Pneumonia

Both Cough and Breathlessness are true.

Therefore:

Pneumonia is derived.

Final conclusions

Fever, Cough, Breathlessness, Flu and Pneumonia

are known for Patient A.

(c) Backward Chaining for Pneumonia

Goal: Patient A has Pneumonia.

Start with:

Pneumonia

Search for a rule that can produce Pneumonia.

Rule 3:

Cough $\wedge$ Breathlessness $\rightarrow$ Pneumonia

Therefore, the goal is reduced into two sub-goals:

Pneumonia

↓

Cough AND Breathlessness

Check the facts:

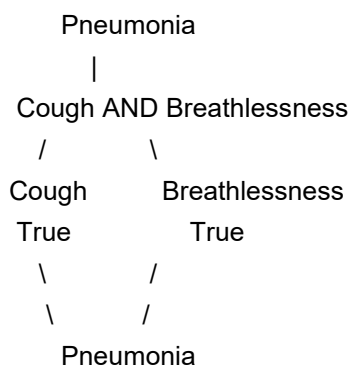
- Cough = True ✓
- Breathlessness = True ✓

Therefore:

Cough $\wedge$ Breathlessness $\rightarrow$ Pneumonia

Hence, the goal is successfully proved.

Goal Reduction Tree



Conclusion: Patient A has Pneumonia.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

(a) Unification

Rule 1

$\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

Facts:

- $\text{Vehicle}(\text{Ambulance})$
- $\text{EmergencyType}(\text{Ambulance})$

We match the variable x with **Ambulance**.

Therefore:

$\theta = \{x / \text{Ambulance}\}$

After substitution:

$\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$

Both conditions are true.

Therefore:

$\text{ClearPath}(\text{Ambulance})$ is derived.

MGU

$\theta = \{x \rightarrow \text{Ambulance}\}$

(b) Resolution Proof for Proceed(CarA)

Rule 1:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

For GreenSignal:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

Rule 3:

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts:

Vehicle(Ambulance)
EmergencyType(Ambulance)
Vehicle(CarA)
Behind(CarA, Ambulance)

Resolution Steps

From Rule 1 and the facts:

Vehicle(Ambulance) + EmergencyType(Ambulance)

we derive:

ClearPath(Ambulance)

From Rule 2:

ClearPath(Ambulance) → GreenSignal(Ambulance)

Therefore:

GreenSignal(Ambulance)

Now use Rule 3:

Vehicle(CarA) ∧ Behind(CarA, Ambulance) ∧ GreenSignal(Ambulance)

All three conditions are true.

Therefore:

Proceed(CarA)

Conclusion

Proceed(CarA) is logically proved.

(c) Two Simultaneous Emergency Vehicles

The current knowledge base can handle multiple vehicles because the rules use variables. However, it does not explicitly provide a rule for handling **two emergency vehicles at the same time**, such as priority ordering or collision prevention.

Additional facts could be:

EmergencyType(Ambulance2)
Vehicle(Ambulance2)

Additional rules could include:

EmergencyType(x) ∧ Vehicle(x) → PriorityVehicle(x)

and:

PriorityVehicle(x) ∧ PriorityVehicle(y) ∧ x ≠ y → ApplyPriorityPolicy(x,y)

This would allow the system to distinguish between multiple emergency vehicles and decide which vehicle should receive priority.

Conclusion: The current KB can identify multiple emergency vehicles, but additional priority and conflict-resolution rules are required for proper simultaneous emergency management.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) **P1: SoilDry → IrrigationNeeded**
- 2) **P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod**
- 3) **P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk**
- 4) **Facts: SoilDry = True, CropWheat = True**

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF).

Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

(a) Conversion to CNF

P1

Given:

SoilDry $\rightarrow$ IrrigationNeeded

Using:

$A \rightarrow B \equiv \neg A \vee B$

Therefore:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

This is already in CNF.

P2

Given:

IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Convert implication:

$\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Using De Morgan's law:

$(\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat}) \vee \text{ApplyDripMethod}$

Therefore:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

P3

Given:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Convert implication:

$\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Using De Morgan's law:

$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Facts

SoilDry = True

CropWheat = True

Therefore, the CNF knowledge base contains:

1. **$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$**
2. **$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$**
3. **$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$**
4. **SoilDry**
5. **CropWheat**

(b) Resolution Refutation for ApplyDripMethod

Goal

ApplyDripMethod

Negate the goal:

$\neg \text{ApplyDripMethod}$

Add this to the KB.

From:

SoilDry

and

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

resolve:

IrrigationNeeded

Now use:

IrrigationNeeded

CropWheat

and:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Resolution gives:

ApplyDripMethod

But we already added:

$\neg \text{ApplyDripMethod}$

Therefore:

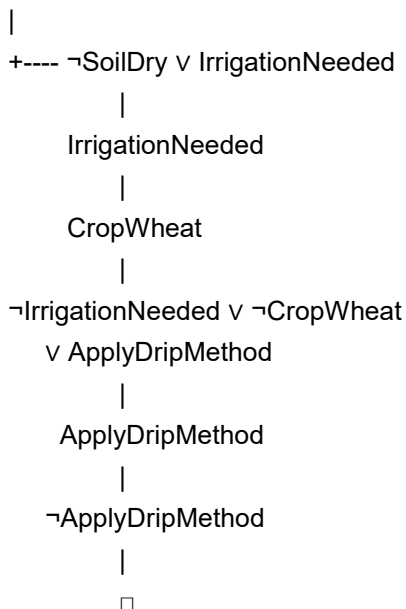
$\text{ApplyDripMethod} \wedge \neg \text{ApplyDripMethod}$

gives the empty clause:

□

Resolution Tree

SoilDry



Conclusion: ApplyDripMethod is logically proved.

(c) Removing SoilDry

Originally:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

When **SoilDry** is removed, there is no fact that directly establishes **IrrigationNeeded**.

Therefore:

IrrigationNeeded cannot be derived.

Since Rule P2 requires:

IrrigationNeeded $\wedge$ CropWheat

we cannot derive:

ApplyDripMethod

Also, P3 requires:

$\neg$ ApplyDripMethod $\wedge$ CropWheat

Although CropWheat is still true, **$\neg$ ApplyDripMethod is not given or derived.**

Therefore:

CropAtRisk cannot be logically concluded.

Final result

Without SoilDry:

- IrrigationNeeded $\rightarrow$ Cannot derive
- ApplyDripMethod $\rightarrow$ Cannot derive
- CropAtRisk $\rightarrow$ Cannot derive

Thus, the removal of SoilDry prevents the reasoning chain from reaching these conclusions.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]**
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]**
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]**
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]**

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

(a) Belief State and Modus Ponens

The initial percepts are:

1. Stench[1,2]
2. Breeze[1,1]
3. Glitter[2,2]

Step 1 — Stench

Rule:

Stench[1,2] $\rightarrow$ Wumpus is adjacent to [1,2]

Since Stench[1,2] is true:

Wumpus is adjacent to [1,2]

is derived.

Step 2 — Breeze

Rule:

Breeze[1,1] $\rightarrow$ Pit is adjacent to [1,1]

Since Breeze[1,1] is true:

Pit is adjacent to [1,1]
is derived.

Step 3 — Glitter

Rule:

$\text{Glitter}[x,y] \rightarrow \text{Gold}[x,y]$

Given:

$\text{Glitter}[2,2]$

Therefore:

$\text{Gold}[2,2]$

is derived.

Belief State

The agent knows:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]
- Wumpus is adjacent to [1,2]
- Pit is adjacent to [1,1]
- Gold is at [2,2]

(b) Forward Chaining for Wumpus and Pit

Wumpus

Given:

$\text{Stench}[1,2]$

Using Rule 1:

$\text{Stench}[1,2] \rightarrow \text{Wumpus adjacent to } [1,2]$

Therefore, the Wumpus can be in a cell adjacent to [1,2].

Possible adjacent cells in the standard grid include:

[1,1], [1,3], [2,2]

subject to the actual world boundaries.

Pit

Given:

$\text{Breeze}[1,1]$

Using Rule 2:

$\text{Breeze}[1,1] \rightarrow \text{Pit adjacent to } [1,1]$

Therefore, the Pit can be in an adjacent cell such as:

[1,2] or [2,1]

depending on the available grid cells.

Gold

Given:

$\text{Glitter}[2,2]$

Using Rule 3:

$\text{Glitter}[2,2] \rightarrow \text{Gold}[2,2]$

Therefore, Gold is definitely at [2,2].

(c) Is the Path [1,1] → [1,2] → [2,2] Safe?

The proposed path is:

[1,1] → [1,2] → [2,2]

At [1,1], the agent perceives Breeze.

This means a Pit may be adjacent to [1,1]. Therefore, moving from [1,1] to [1,2] has a possible danger.

At [1,2], the agent perceives Stench.

This means a Wumpus may be adjacent to [1,2]. Therefore, [1,2] cannot be confirmed as safe from the given knowledge.

At [2,2], Glitter is perceived, so:

Gold[2,2]

is known.

However, the given KB does not establish:

¬Wumpus[1,2]

or

¬Pit[1,2]

Therefore, the safety rule cannot prove that [1,2] is safe.

Conclusion

The path cannot be guaranteed safe using the available knowledge. The agent should not assume that [1,2] is safe because both the Wumpus and Pit locations have possible dangerous alternatives.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand